

对映选择性分析：在二乙基锌与苯甲醛不对称 加成中手性配体设计的逻辑推理

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摘要 基于二乙基锌与苯甲醛不对称加成反应中催化剂的构象和对映选择性关系的数学表达式, 通过对手性催化剂的动态构象行为的对映选择性分析, 详细描述了一个理性设计手性配体的方法, 即从一个最简单的只含有一个手性中心的 β -氨基醇出发, 通过对催化剂动态构象行为的合理控制, 理性设计出几乎是所有报道的具有高对映选择性的 β -氨基醇类手性配体, 共 94 个。

关键词 对映选择性分析; 构象控制; 理性设计; 手性配体; 二乙基锌

Enantioselective Analysis: Logic of Chiral Ligand Design for Asymmetric Addition of Diethylzinc to Benzaldehyde

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Abstract A detailed logic-guided approach towards chiral ligands design is described via the enantioselective analysis of the dynamic conformational behaviors of catalyst, which is based on a mathematical relationship between conformations and enantioselectivity, for asymmetric addition of diethylzinc to benzaldehyde. Following this logic thought, 94 examples, almost all highly enantioselective β -aminoalcohol ligands reported, can be rationally devised by the logic control of the dynamic conformational behaviors of the catalyst from the simplest β -aminoalcohol with one single chiral center as starting point.

Keywords enantioselective analysis; conformational control; rational design; chiral ligand; diethylzinc

1 Introduction

Over the past several decades, tremendous effort has been dedicated to the development of the chiral ligands and catalysts, which has been a principal driving force in the rapid advance of the field of asymmetric catalysis. Because of a clear lack of rational and systematic guidelines for chiral ligand and catalyst design, asymmetric catalyst discovery relies mainly on extensive trial and error strategy based on chemical intuition and empiricism. As pointed out by Knowles,^[1] “Since achieving 95% *ee* only involves energy differences of about 8.4 kJ, which is no more than the barrier encountered in a simple rotation of ethane, it is unlikely that before the fact one can predict what kind of ligand structures will be effective.” For this unpredictable reason, modifiable and

modular ligand backbones become a key to solving the long-standing challenge of ligand/catalyst development. In practice, in order to adapt the steric and electronic effect of the ligand to the unique requirements of an asymmetric transformation, a large number of the candidates of chiral ligands are synthesized, and subsequently screened in terms of the evaluation of activity and enantioselectivity. As a result, finding an optimal chiral ligand usually not only occupies time and minds of many investigators, but also requires a large investment of manpower and funding, severely hampering the optimization of the asymmetric process.

However, attempts to the rational design of chiral ligands never cease, such as computation chemistry^[2] and mechanistic knowledge/principles.^[3] Unfortunately, to the best of our knowledge, a simple, practical, general strategy for the evaluation and prediction of the enanti-

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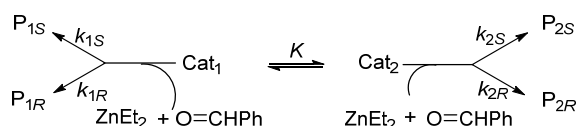
Dedicated to Professor Yangjie Wu on the occasion of his 90th birthday.

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oselectivity has not been reported to date.

For a long time, we have been exploring some main structural factors affecting enantioselectivity of chiral ligands in asymmetric addition of organozinc to aldehydes.^[4a~4l] Recently, we discovered that the necessary relationship existed between the relative amounts of free ligand's conformations and the experimental enantioselectivity in asymmetric ethylation of benzaldehyde.^[4m] In addition, this phenomenon was further demonstrated by subsequent theoretical calculation and experiment. Based on this principle, we described the logic design of a pair of chiral diastereomeric ligands exhibiting nearly the same excellent enantioselectivity in the asymmetric addition of diethylzinc to aldehydes from a single common chiral source.^[4n] In order to provide a theoretical support for our previous studies on the relationship between conformation and enantioselectivity, we established the first catalytic asymmetric kinetic system for a flexible chiral catalyst (Scheme 1), and more importantly, a quantitative relationship between conformations and enantioselectivity was derived from this catalytic asymmetric kinetic system (Eq. 1).^[4o]



Scheme 1 Catalytic asymmetric kinetic system for a flexible chiral catalyst

Cat₁ and Cat₂ are two conformers of a chiral catalyst that is formed by the reaction of a diethylzinc with a chiral β -amino alcohol

In order to provide a simple, practical and efficient method for the design of chiral ligands/catalysts (relative to trial and error method), and simultaneously demonstrate reliability and generality of this mathematical relationship, in this manuscript, we described a detailed logic-guided approach towards chiral ligands design via the enantioselective analysis of the dynamic conformational properties and behaviors of catalyst. Our results indicated that the enantioselectivity of chiral ligands and catalysts could be predicted and chiral ligands/catalysts could be designed rationally.

1 Results and discussion

The theoretical basis for the logical design of chiral ligands and catalysts is a quantitative relationship between conformations and enantioselectivity (Eq. 1), which is derived from the catalytic asymmetric kinetic system for a flexible chiral catalyst (Scheme 1).^[4o]

$$ee = \frac{x_1(k_{1R} - k_{1S}) + x_2(k_{2R} - k_{2S})}{x_1k_1 + x_2k_2} \quad (1)$$

where (*R*)-enantiomer is the major product of the catalyzed reaction, (*S*)-enantiomer is the minor one. The term x_i and k_i refer to the mole fraction and reaction rate constant, respectively, of the i th conformer of the chiral

catalyst, *i.e.* $x_i = [\text{Cat}_i] / \sum [\text{Cat}_i]$, $k_i = k_{iS} + k_{iR}$. The quantity k_{iR} and k_{iS} , respectively, are reaction rate constants of (*R*)- and (*S*)-enantiomers produced by the i th conformer catalysis.

Eq. 1 is transformed into its mathematical equivalent Eq. 2.

$$ee = \frac{x_1(k_{1R} - k_{1S})}{k_{W-H}} + \frac{x_2(k_{2R} - k_{2S})}{k_{W-H}} \quad (2)$$

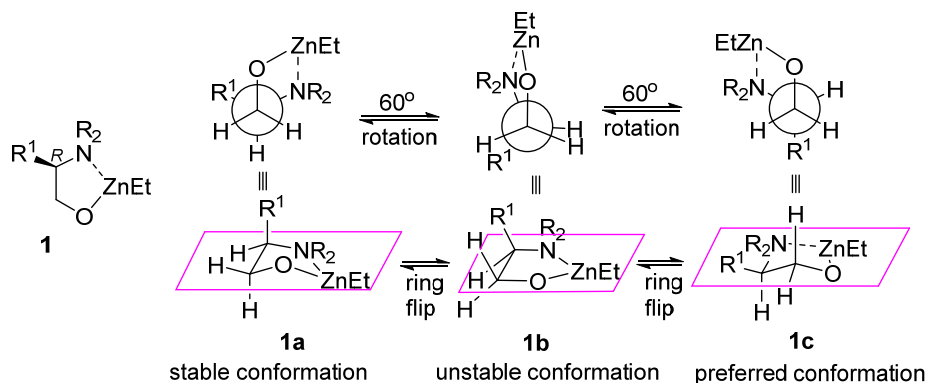
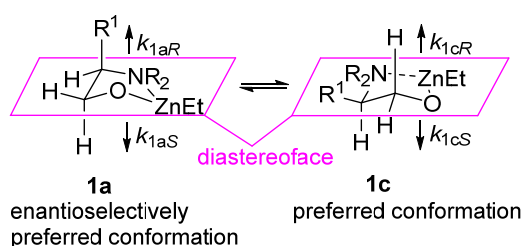
where the term k_{W-H} is defined to be the Winstein-Holness reaction rate constant, namely, $k_{W-H} = \sum x_i k_i$ (where $k_{W-H} = x_1k_1 + x_2k_2$).^[5]

This quantitative expression shows that all conformational isomers of a catalyst make the contribution to the observed enantioselectivity. In addition, the term $x_i(k_{iR} - k_{iS})/k_{W-H}$ is viewed as the contribution of the i th conformational isomer to the total enantioselectivity. In order to exemplify the logic of the design of chiral ligand/catalyst based on the quantitative expression Eq. 2, asymmetric addition of diethylzinc to benzaldehyde catalyzed by β -aminoalcohols is selected for study^[6] because (1) the observed enantioselectivity is very sensitive to the subtle changes in the chiral ligand's structure; (2) almost all conceivable backbones of β -aminoalcohols have been designed and synthesized to induce this asymmetry; (3) the structure of the catalysts, one five-membered zinc ring, is relatively simple.

The previous mechanism investigation by Noyori *et al.*^[7] indicated that the five-membered zinc aminoalkoxide **1** is believed to be a bifunctional catalyst because the complex **1** acts as a Lewis acid to activate carbonyl substrates and also as a Lewis base to activate diethylzinc reagents. Both aldehyde and diethylzinc can coordinate either face of the five-membered chelate ring, which is called a diastereoface and then undergo *anti*-*ti*-configured 5/4/4 tricyclic transition state, to give a pair of unequal amounts of (*R*)- or (*S*)-enantiomers with the rate constants k_R and k_S , respectively. The relative magnitudes of the rate constants k_R and k_S are mainly dependent on the 1,2- and 1,3-nonbonded interactions between the groups on α - and β -carbon atoms and the entering substrate and reagent during coordination process since alkyl groups have relatively minor electronic effects, which are usually unrelated to the size of the alkyl group itself. The less hindered side of the five-membered zinc ring shows faster rate (*i.e.*, the k value is large), vice versa.

Newman projection formula clearly shows that the flexible catalyst **1** has two stable gauche conformations **1a** and **1c**, as well as a very unstable eclipsed conformation **1b** because of carbon-carbon single bond rotation (Scheme 2).

Theoretical computation further demonstrates that there are two stable gauche conformations **1a** and **1c** in five-membered Zn-chelate ring **1**.^[8] For simplicity, two gauche conformers **1a** and **1c** are only considered in a conformational equilibrium of the catalyst **1** (Scheme

Scheme 2 Conformational analysis of chiral catalyst **1**

Scheme 3 Enantioselective analysis of dynamic conformational behavior

The quantities k_{iR} and k_{iS} are reaction rate constants of (*R*)- and (*S*)-enantiomers produced by the *i*th conformer catalysis, respectively

3). The steady CD signals of very stable 1 : 1 complexes, which are formed by the reaction of chiral β -aminoalcohols and Lanthanide chelates, clearly confirm that the gauche conformation **1c** with a bulky group in equatorial position is favored over the gauche arrangement **1a** with a large substituent in axial position (Scheme 3).^[9] This means that the presence of the equatorial group stabilizes the catalyst conformation **1c**, called the preferred conformation. The similar conformational behavior is also revealed in five-membered hydrogen-chelate ring of chiral β -aminoalcohols.^[10]

The diastereoface of the five-membered zinc cycle in **1a** has a very large differentiation due to the different steric hindrance at above and under sides of the zinc ring. As indicated in Scheme 3, the conformer **1a** has a large steric effect at the k_{1aR} face as compared with the k_{1aS} side since the existence of a large axial substituent R^1 in **1a** can effectively block the approach of reactants from the top face of the five-membered zinc ring, i.e., $k_{1aR} \ll k_{1aS}$. This means that the catalyst conformation **1a** has high diastereofacial selectivity, leading to high enantioselectivity [i.e. $ee_{1a} = (k_{1aS} - k_{1aR}) / (k_{1aS} + k_{1aR})$].^[40] Contrarily, the catalyst conformation **1c** has a very small differentiating diastereoface owing to the similar steric hindrance at k_{1cS} and k_{1cR} sides of the zinc cycle, respectively, resulting in low enantioselectivity [i.e. $ee_{1c} = (k_{1cS} - k_{1cR}) / (k_{1cS} + k_{1cR})$].^[40] Generally, the degree of the diastereofacial differentiation (or level of diastereofacial selectivity) mainly depends on the magnitude of the rate

constant difference such as $k_{1aS} - k_{1aR}$. In addition, the big difference of rate constant (or the large diastereofacial differentiation) will lead to high enantioselectivity since the *ee* value is directly proportional to the rate constant difference [e.g. $ee_{1a} = (k_{1aS} - k_{1aR}) / (k_{1aS} + k_{1aR})$]. The catalyst conformation **1a** is called the enantioselectively preferred conformation because the conformer **1a** shows the much higher level of enantioselectivity than **1c**.

Following Eq. 2, the total enantioselectivity ee_1 of the chiral catalyst **1** is expressed as Eq. 3.

$$ee_1 = \frac{(k_{1aS} - k_{1aR})}{k_{1W-H}} x_{1a} + \frac{(k_{1cS} - k_{1cR})}{k_{1W-H}} x_{1c} \quad (3)$$

where $k_{1W-H} = x_{1a}k_{1a} + x_{1c}k_{1c}$, $k_{1a} = k_{1aS} + k_{1aR}$, $k_{1c} = k_{1cS} + k_{1cR}$, $x_{1a} + x_{1c} = 1$.

As indicated in Eq. 3, the *ee* value of the flexible catalyst **1** principally depends on the contributions of its conformers **1a** and **1c**. In addition, the contribution of each catalyst conformation to the observed enantioselectivity is relevant to their respective mole fraction and diastereofacial differentiation. The contributing conformer **1a** has a large diastereofacial differentiation because of $(k_{1aS} - k_{1aR}) \gg (k_{1cS} - k_{1cR})$, but its relative populations (or its mole fraction) are very small in the conformation equilibrium owing to a bulky substituent R^1 in the axial position. This implies that the conformation **1a** affords a low contribution to the enantioselectivity, that is, the quantity $(k_{1aS} - k_{1aR})x_{1a}/k_{1W-H}$ is small. Similarly, the conformational contributor **1c** exists in significant populations because the substituent R^1 in **1c** is equatorial, yet the conformer **1c** has such poor diastereodiscriminating ability that it also makes an insignificant contribution to the enantioselectivity. The mentioned-above two factors, combined with Eq. 3, lead to the low enantioselectivity of the catalyst **1**. In fact, when R^1 is a methyl group, the observed enantioselectivity is only 33% *ee* (Figure 1, **1-1**). Even if R^1 is a very large bulky group *t*-Bu, the *ee* values of ligands **1**~**4** are only 36%~56%.

Those disadvantages of the chiral catalyst **1** are

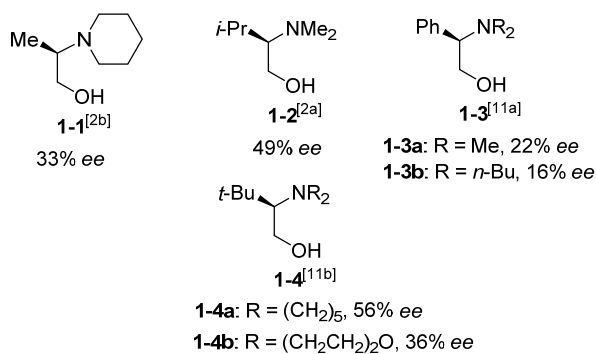
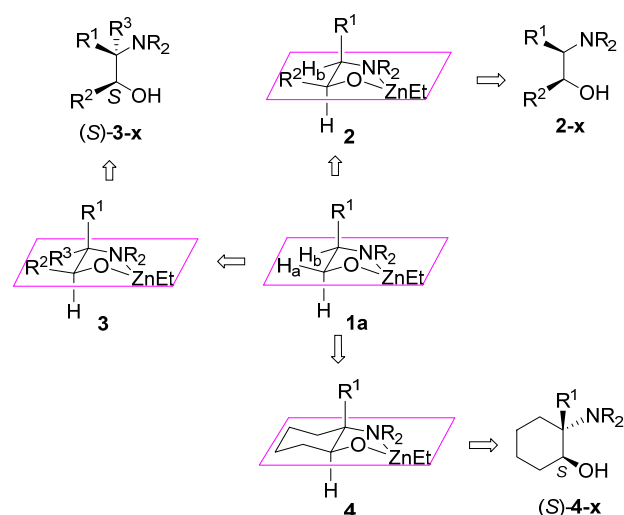


Figure 1 ee values of chiral ligands reported

overcome by appropriate introduction of substituents into the carbon backbone of conformers, either **1a** or **1c**, even **1b** since suitable substitution patterns not only regulate equilibrium ratios of conformation, but also control spatial orientation of substituents in favor of a certain special conformer, which is optimal for enantioselectivity. In principle, according to mathematical relationship Eq. 3, ideal scenarios are: (1) the increase in the relative populations of the enantioselectively preferred conformation **1a** via appropriate replacement patterns leads to the enhancement of the ee value, whilst holding the diastereofacial differentiation ($k_{1a\ S} - k_{1a\ R}$) constant;^[40] (2) the augment of diastereofacial differentiation of the preferred conformation **1c** by suitable incorporation of substituents results in the improvement of the total enantioselectivity when the contribution of **1a** is kept invariant;^[4p] (3) a combination of (1) and (2), *i.e.*, the increase in the relative populations of a certain special conformation, and the simultaneous improvement of its diastereofacial differentiation realize the enhancement of the overall enantioselectivity. These strategies are the theoretical basis for rational design of chiral ligands and catalysts.

In order to exemplify the logical thought of chiral ligand design, the desired conformation **1a** is selected as a starting point for rationally devising highly enantioselective ligands. A logical analysis process for chiral ligands **2-x**, (*S*)-**3-x**, and **4-x** is illustrated in Scheme 4.

The conformer **1a** has a relatively low concentration in the conformational equilibrium owing to the presence of a bulky axial substituent R¹. In order to increase the relative population of **1a**, the substitution of the equatorial position H_a in **1a** with a large bulky substituent R² gives diastereomer **2**. With respect to the diastereoface of the five-membered zinc ring of diastereomers **1a** and **2**, respectively, the parent conformation **1a** has the same diastereofacial differentiation as the isomer **2** because of their sterically equivalent hindrance at above and under sides of the zinc ring, respectively. Diastereomers **1a** and **2** are referred to as the enantioselectively equivalent conformation. However, the relative proportions of **2** will be increased in the redistribution of the conformational equilibrium compared with the parent **1a** since the



Scheme 4 Logical analysis process for highly enantioselective ligands **2-x**, (*S*)-**3-x** and **4-x**

introduction of the equatorial substituent R² stabilizes more conformer **2** relative to **1a**. Following Eq. 3, the chiral catalysts **2** will show very significant enhancement in the enantioselectivity relative to the corresponding ones **1**. Actually, the comparison of chiral ligands **2-x** (Figure 2) with **1-x** (Figure 1) shows that the introduction of the substituent R² leads to substantial improvement in ee values (**1-1** vs. **2-1**, **2-2**, **2-3**, **2-4**, **2-5** and **2-6**; **1-2** vs. **2-7**, **2-8**, **2-9**, **2-10**; **1-3** vs. **2-11**, **2-12**, **2-13**, **2-14**, **2-15** and **2-16**, respectively).

The position of the new conformational equilibrium is primarily dependent on the size of the equatorial R². The larger is the bulkiness of the group R², the more is the relative population of the conformer **2** in new equilibrated state. When R² is *t*-Bu group, the relative amounts of **2** reach a maximum value.

In the same way, the simultaneous replacement of both equatorial H_a and H_b in **1a** by the R² and R³ groups, respectively, affords the trisubstituted derivative **3** with the equivalent diastereofacial differentiation as compared with **1a** and **2** (Scheme 4). In addition, the diastereomer **3** should possess a higher conformational population in new conformational equilibrium than **2** due to two bulky substituents R² and R³ of **3** in the equatorial position. According to Eq. 3, the enantioselectivity of chiral catalysts **3** is further enhanced. In fact, the chiral ligands **3-x** (Figure 3) exhibit the trend of more enantioselectivity than **2** [**2-12** vs. (*S*)-**3-3**; **2-16** vs. (*S*)-**3-1** and (*S*)-**3-2**, respectively].

Tethering both equatorial H_a and H_b in **1a** with a chain gives a ring fusion derivative **4**. This means that the enantioselectively preferred conformation **1a** can be completely fixed by ring annellation. The fully rigid conformation **4** with a *trans*-fused bicyclo[4.3.0]nonane ring system has a maximum population, and simultaneously the same diastereofacial differentiation as isomer **1a** is maintained. Hence the catalyst **4** should lead to the

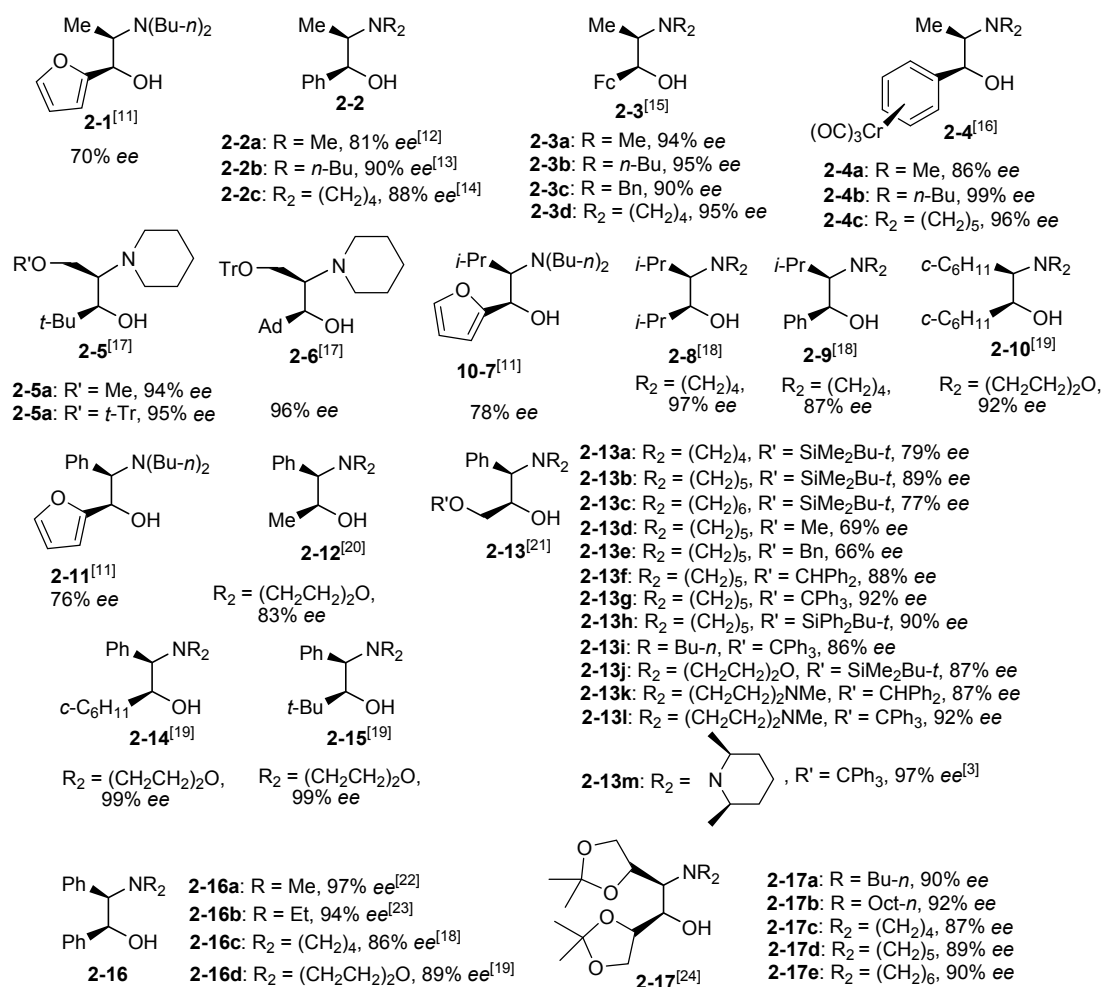


Figure 2 ee values of chiral ligands reported

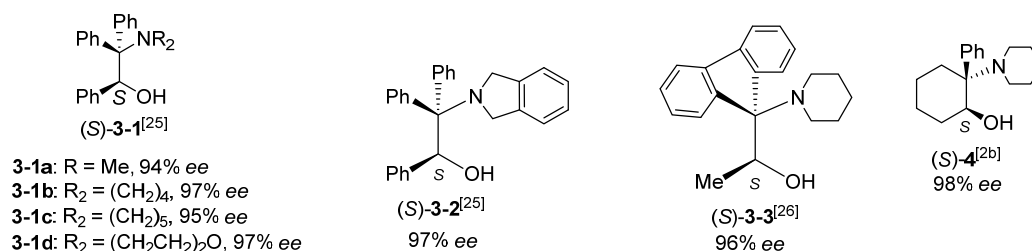


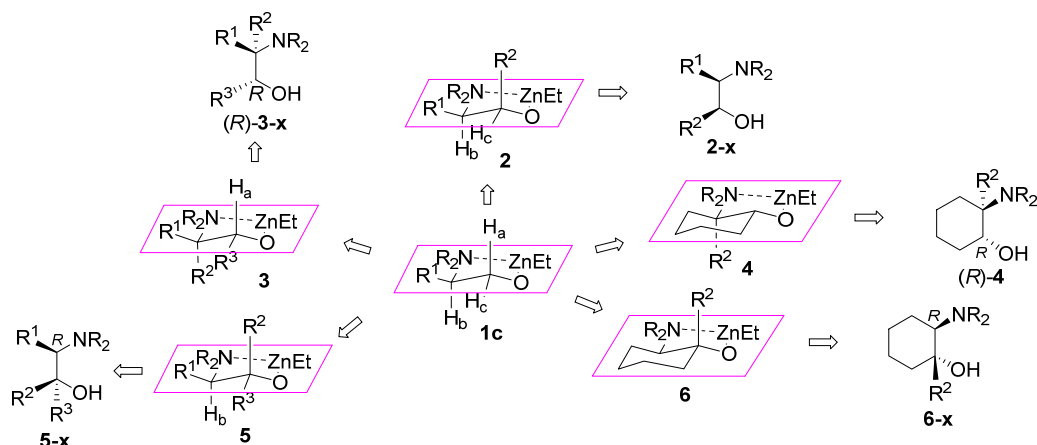
Figure 3 ee values of chiral ligands reported

outstanding enantioselectivity. Actually, the chiral ligand (*S*)-**4** shows superior level of enantioselectivity of up to 98% ee (Figure 3).

The preferred conformation **1c** is also considered as a beginning point that can attain chiral ligand skeletons with high enantioselectivity (Scheme 5).

The diastereoface of the five-membered zinc cycle in **1c** has a very small diastereofacial differentiation owing to the similar steric hindrance between top and bottom faces of the zinc ring. In order to increase the diastereofacial discriminating ability of **1c**, following strategies are used to enhance the reaction enantioselectivity: (1) The substitution of the axial H_a in the preferred confor-

mation **1c** generates a conformer **2** possessing a large differentiating diastereoface, from which high enantioselective ligands **2-x** are derived (Figure 2). In order to avoid decreasing in relative populations of **2**, which are caused by the introduction of the axial R², the displacement of the axial H_a in **1c** by R² and the simultaneous replacement of the equatorial H_c by R³ produce the conformational isomer **5**, offering chiral ligands **5-x** (Figure 4) with higher enantioselectivity than the corresponding **2-x** (Figure 2). (2) For the same reason, chiral ligands (*R*)-**3-x** with very high ee values as enantiomers of (*S*)-**3-x** (Figure 3) are derived when both axial H_b and equatorial H_c in **1c** are replaced with bulky groups are



Scheme 5 Logical analysis process for highly enantioselective ligands **2-x**, **(R)-3-x**, **(R)-4-x**, **5-x**, and **6-x**

are replaced with bulky groups R^2 and R^3 , respectively. (3) Either axial H_a in **1c** or axial H_b is substituted by a bulky group R^2 , and simultaneously the equatorial H_c and vicinal *anti*-group R^1 chain bonds are both incorporated into a chair six-membered ring. As a result, the completely rigid chiral catalysts **6** and **4**, respectively, are afforded, meaning that this structural modification not only enhance diastereofacial differentiation of **1c**, but also attain the maximum populations. Chiral ligands **(R)-4**, as the enantiomer of **(S)-4** (Figure 3), and **6-x** (Figure 5) are logically designed, respectively. For the

fully rigid catalysts **4** and **6** with a *trans*-fused bicyclo-[4.3.0]nonane ring, even if there are axial substituents at 4- or 5-position on the chair six-membered ring, they have almost no influence on asymmetric induction taking place at the active site of the reaction because those groups are far removed from the metal site. As a result, the additional chiral centre in **6-3** results in the slight decrease in ee values when compared with chiral ligands **6-1** (**6-3c** vs. **6-3a**, **6-3e** vs. **6-1b**).

Although the eclipsed conformation **1b** is a very unstable, it can be used as an entry point for the rational

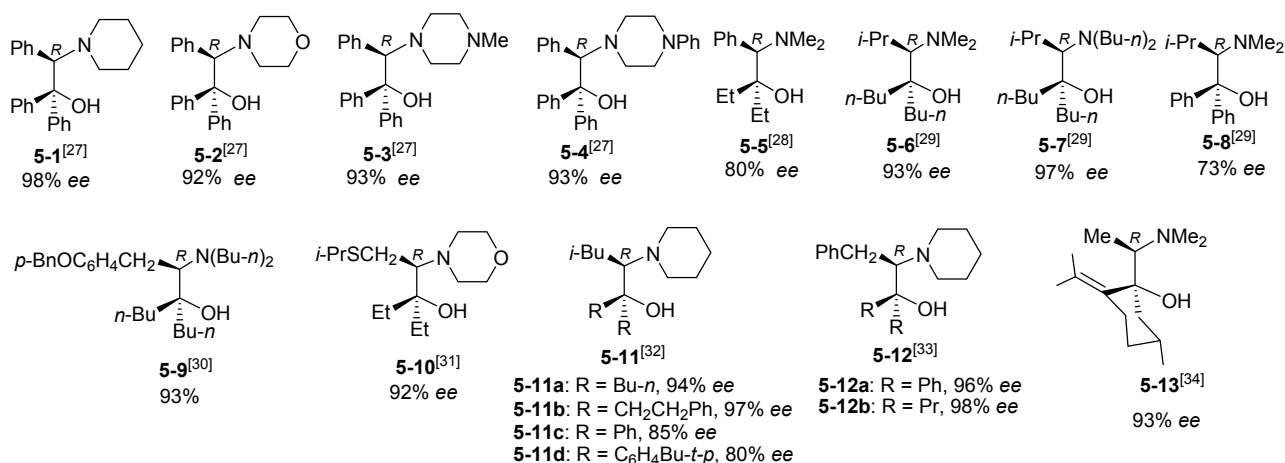


Figure 4 ee values of chiral ligands reported

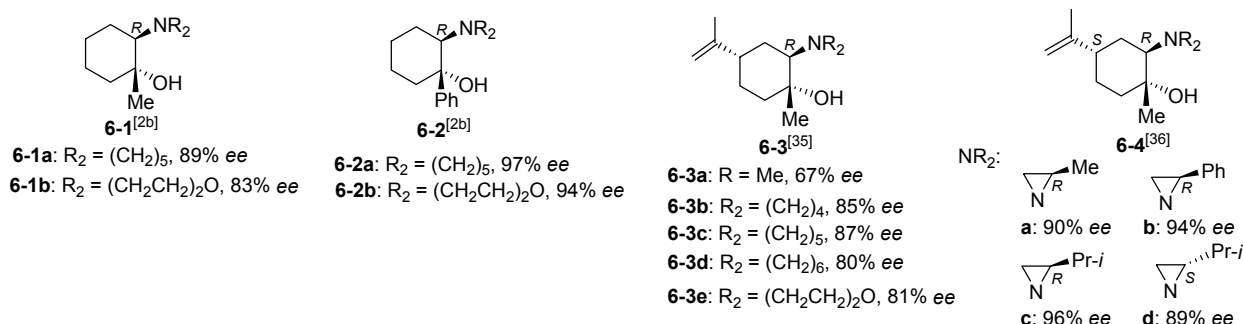


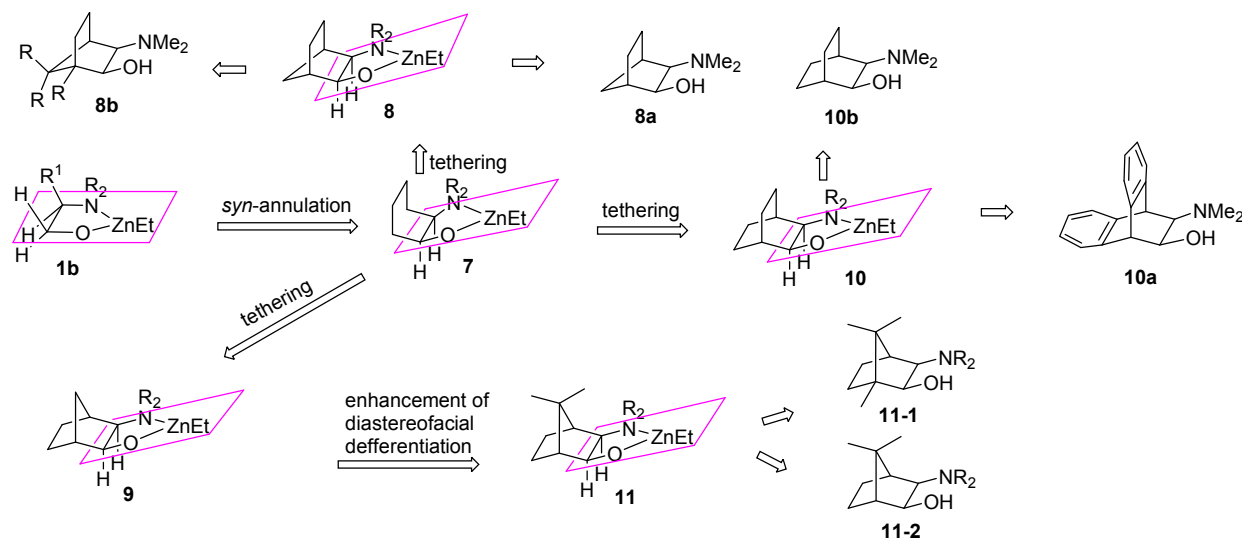
Figure 5 ee values of chiral ligands reported

design of chiral ligands by means of described-above strategies as well (Scheme 6). In order to maintain the spatial disposition of eclipsed conformation **1b**, the incorporation of the substituent R^1 and vicinal *syn*-hydrogen chain bonds into a boat six-membered ring, as in **7**, being very unstable, and the simultaneous locking of the unstable boat conformation **7** by another short bridges, furnishes fully rigidly eclipsed conformations such as **8**, **9** and **10**. Chiral ligand **8a** with enantioselectivity of up to 94% *ee* (Figure 6) is derived from **8**. The R groups on chiral ligand **8b** ($R=Me$), readily synthesized from natural product camphor, do not almost influence the diastereofacial differentiation of the rigid skeleton since *gem*-dimethyl substituents is far from active sites of chemistry reaction, and the bridge-head methyl group is nearly coplanar with the five-membered zinc ring, so **8a** and **8b** are enantioselectively equivalent diastereomers. In deed, chiral ligands **8a** and **8b** show nearly the same outstanding enantioselectivity (Figure 6).

To further increase the diastereofacial differentiation of **9**, the introduction of germinal dimethyl groups into the catalyst **9**, as in **11**, enhances the reaction enantioselectivity. Actually, the diastereomers **11-1** exhibit the outstanding enantioselectivity of up to 98%~99% *ee* (Figure 6), which is easily prepared from chiral natural product camphor. Probably because of difficult synthesis, the chiral ligands **11-2** as enantioselectively equivalent diastereomers of **11-1** has not been reported so far (Scheme 6).

The rigid chiral ligands **8a** and **10b** are enantioselectively equivalent diastereomers due to their equal diastereofacial differentiation. However, the ligand **10b** has not been reported up to now for possible reason that the enantiopure **10b** is difficult to be synthesized by asymmetric transformation. Contrarily, the chiral ligand **10a** is readily made through asymmetric synthesis. In addition, the chiral ligand **10a** should show more enantioselectivity than **10b** because **10a** has a larger differentiating distereoface relative **10b**. In fact, the chiral ligand **10a** affords elevated levels of asymmetric induction (96% *ee*) (Figure 6).

The very unstable coat conformation **7** spontaneously converts the stable chair ones **12** and **13** via ring flip (Scheme 7), which are diastereomers with a stereodifferentially equivalent diastereoface. In order to guarantee the spatial orientation of chair conformations **12** and **13**, and simultaneously augment their diastereodifferentiating ability, the fixation of diastereomers **12** and **13** with short tethers provides completely rigid chair conformations such as **14**, **15**, **16**, and **17**. To further increases diastereofacial differentiation of **16** and **17**, the introduction of *germinal* dimethyl groups into the catalyst **16** and **17** gives catalysts **18** and **19**, respectively. The rigid chiral catalysts **18** and **19** are diastereomers, but they should show the same catalytic behaviors due to equal rigidity and equivalent diastereofacial differentiation. In fact, chiral ligands (1*S*,2*S*,3*R*,5*R*)-**18**, and (1*R*,2*R*,3*S*,5*S*)-**19** as enantiomers of (1*S*,2*S*,3*R*,5*R*)-**19** exhibit the same enantioselectivity of up



Scheme 6 Logical analysis process for highly enantioselective ligands **8x**, **10x**, and **11-x**

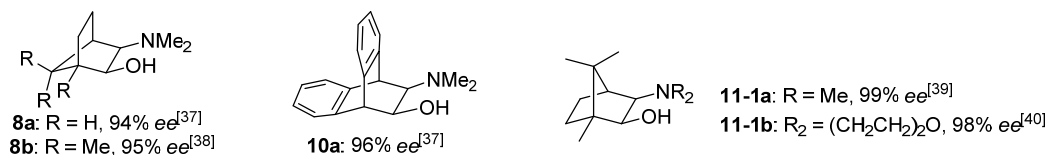


Figure 6 *ee* values of chiral ligands reported

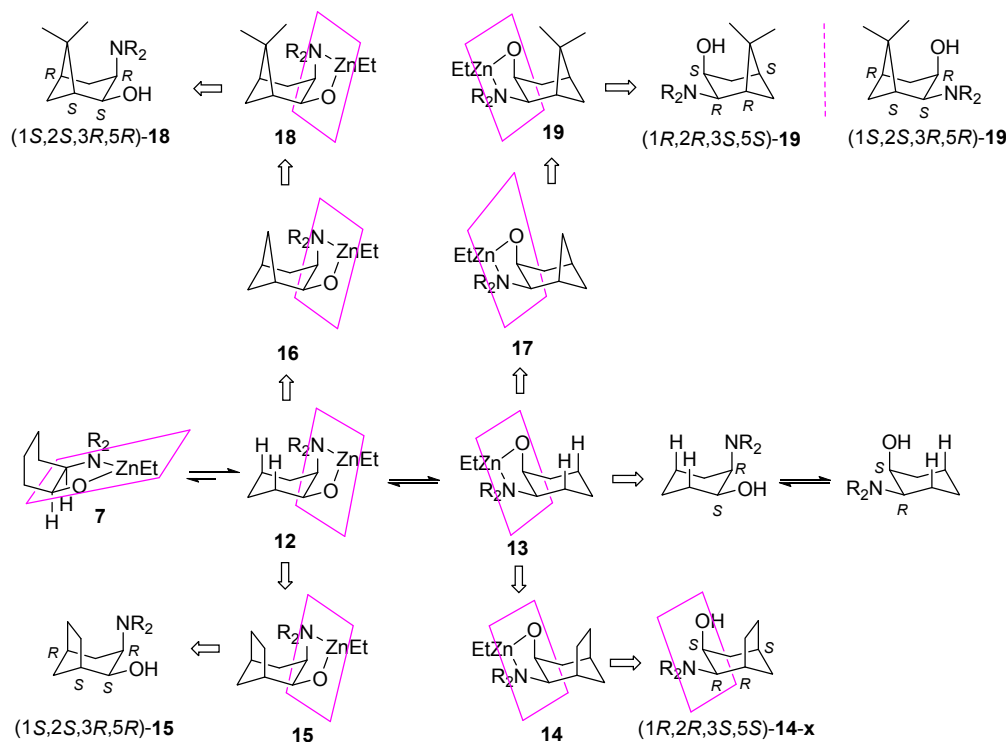
to 99% *ee* in $-15 \sim 0$ °C (Figure 7).

For the same reason, the catalysts **14** and **15** are also enantioselectively equivalent diastereomers with a large differentiating diastereoface. In fact, chiral ligands (1*R*,2*R*,3*S*,5*S*)-**14-1** and (1*S*,2*S*,3*S*,5*R*)-**14-2** exhibit more than 95% *ee*, which is derived from chiral catalyst **14** (Figure 7).

2 Conclusions

In summary, we have established a general and systematic logic analysis procedure for the design of chiral ligands for asymmetric addition of diethylzinc to benzaldehyde, which includes tree strategies: (1) the relative populations of the enantioselectively preferred

conformation are increased in the conformational equilibrium of the catalyst via appropriate replacement patterns; (2) the diastereofacial differentiation of the preferred conformation is augmented by suitable incorporation of substituents; (3) a combination of (1) and (2), *i.e.*, the relative populations of a certain special conformation of the catalyst are enhanced, simultaneously, its diastereofacial differentiation is improved. In addition, this approach can be used to not only rationally devise chiral ligands/catalysts, but also correctly predict and assess the reaction enantioselectivity. Studies are underway further using the new concept of enantioselective analysis to rationally design chiral ligands for other asymmetric catalysis.



Scheme 7 Logical analysis process for highly enantioselective ligands **18x**, **19** and **14-x**

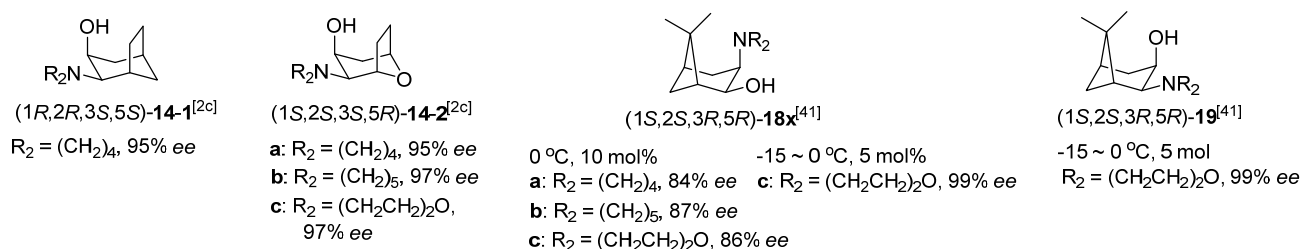


Figure 7 *ee* values of chiral ligands reported

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